

Trainings ▾

Preparatory Steps

Automotive with CM875, CM870 and CM570

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the battery. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Remove the air intake piping from the air cleaner to the turbocharger compressor inlet. Refer to the OEM service manual.
- Remove the charge-air cooler piping from the air intake adapter to the turbocharger compressor outlet. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-027-tr.html>)

- Remove the air compressor inlet tube from the EGR mixer. Refer to Procedure 012-109 in Section 12.
(/qs3/pubsys2/xml/en/procedures/35/35-012-109-tr.html)
- Remove the EGR mixer. Refer to Procedure 011-021 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-021-tr.html)
- Disconnect the EGR temperature sensor and intake manifold air temperature sensor connectors from the engine harness.
- Remove the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html)
- Remove the EGR connection tube. Refer to Procedure 011-025 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html)
- Remove the air intake connection. Refer to Procedure 010-080 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-080-tr.html)
- Remove the intake manifold pressure sensor. Refer to Procedure 019-061 in Section 19.
(/qs3/pubsys2/xml/en/procedures/82/82-019-061.html)
- Remove the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html)
- Remove the exhaust gas pressure sensor. Refer to Procedure 019-376 in Section 19
(/qs3/pubsys2/xml/en/procedures/16/16-019-376.html)
- Remove the EGR differential pressure sensor. Refer to Procedure 019-370 in Section 19.
(/qs3/pubsys2/xml/en/procedures/16/16-019-370.html)
- Remove the EGR differential pressure sensor adapter. Refer to Procedure 011-028 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-028-tr.html)
- Remove the EGR valve coolant lines. Refer to Procedure 011-030 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-030-tr.html)
- Remove the coolant vent line. Refer to Procedure 008-017 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-

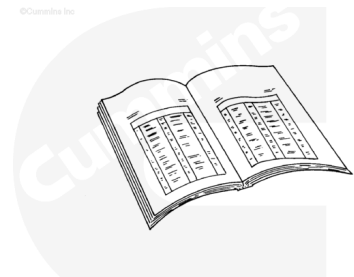
008-017-tr.html)

- Remove the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)
- Remove the EGR valve. Refer to Procedure 011-022 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html)
- Remove the water manifold and return tube. Refer to Procedure 008-061 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html)
- Remove the EGR cooler. Refer to Procedure 011-019 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-019-tr.html)
- Remove the lifting bracket. Refer to Procedure 016-001 In Section 16. (/qs3/pubsys2/xml/en/procedures/35/35-016-001-tr.html)

Automotive with CM570

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



ck800wa

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⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Drain the coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)

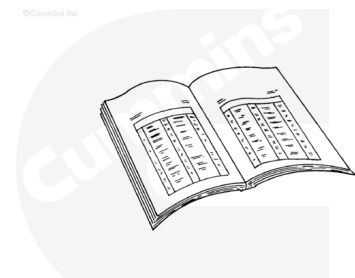
- Disconnect the battery. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Disconnect the air pipe from the charge-air cooler to the inlet air connection. Refer to the OEM service manual.
- Remove the inlet air connection and the sensors from the air manifold.

Automotive with CM876

⚠ WARNING ⚠

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Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Note : Brush away any loose dirt from around the area of the air handling connections to avoid contamination of the interior of the engine.

- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Remove the air intake piping from the air cleaner to the turbocharger compressor inlet. Refer to the OEM service manual. Cover any open points with caps from the Cummins® Vehicle Air Plumbing Clean Care Kit.
- Remove the charge-air cooler piping from the air intake adapter to the turbocharger compressor outlet. Cover any

open points with caps from the Air Handling Clean Care Kit.

- Remove the air compressor inlet tube from the EGR mixer. Refer to Procedure 012-109 in Section 12. (</qs3/pubsys2/xml/en/procedures/35/35-012-109-tr.html>)
- Remove the EGR mixer. Refer to Procedure 011-021 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-021-tr.html>)
- Disconnect the EGR temperature sensor and intake manifold air pressure / temperature sensor connectors from the engine harness.
- Remove the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html>)
- Remove the EGR connection tube. Refer to Procedure 011-025 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html>)
- Remove the air intake connection. Refer to Procedure 010-080 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-080-tr.html>)
- Remove the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html>)
- Remove the exhaust gas pressure sensor. Refer to Procedure 019-376 in Section 19. (</qs3/pubsys2/xml/en/procedures/16/16-019-376.html>)
- Remove the EGR differential pressure sensor. Refer to Procedure 019-370 in Section 19. (</qs3/pubsys2/xml/en/procedures/16/16-019-370.html>)
- Remove the EGR differential pressure sensor adapter. Refer to Procedure 011-028 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-028-tr.html>)
- Remove the turbocharger and turbocharger actuator coolant return lines. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-041-tr.html>)
- Remove the coolant vent line. Refer to Procedure 008-017 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-017-tr.html>)

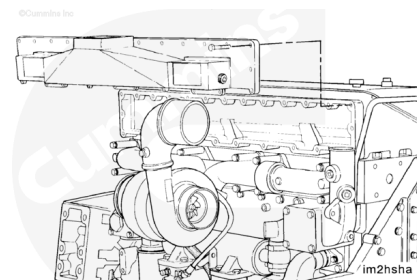
- Remove the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)
- Remove the EGR valve and mounting bracket. Refer to Procedure 011-022 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html)
- Remove the water manifold and return tube. Refer to Procedure 008-061 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html)
- Remove the aftertreatment fuel injector coolant return line. Refer to Procedure 011-051 in Section 11.
(/qs3/pubsys2/xml/en/procedures/101/101-011-051-tr.html)
- Remove the EGR cooler. Refer to Procedure 011-019 In Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-019-tr.html)
- Remove the lifting bracket. Refer to Procedure 016-001 in Section 16. (/qs3/pubsys2/xml/en/procedures/35/35-016-001-tr.html)

Remove

Automotive with CM875, CM870 and CM570

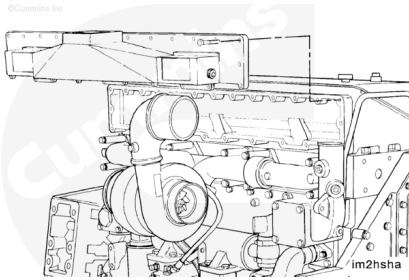
The air intake manifold has five different size capscrews. When removing the capscrews, note their location. It can be helpful to separate the capscrews according to size.

Remove the air intake manifold capscrews and manifold.



Automotive with CM570

Remove the mounting capscrews and the air intake manifold.



Automotive with CM876

The air intake manifold has six different size capscrews.

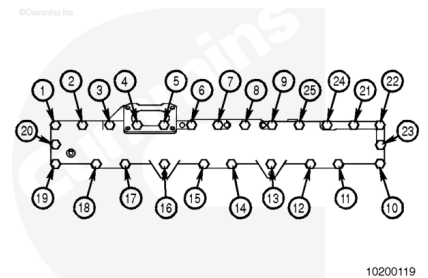
When removing the capscrews, note their location. It can be helpful to separate the capscrews according to size.

Two special stainless steel capscrews, labeled number 4 and 5, are used for corrosion resistance.

Make sure these capscrews are installed at these locations during assembly.

Remove the air intake manifold capscrews, EGR valve mounting adapter, and air intake manifold.

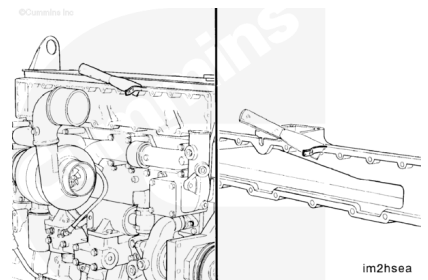
Cover open points in the air flow passages of the intake cover and rocker cover with heavy tape.



Clean and Inspect for Reuse

Automotive with CM875, CM870 and CM570

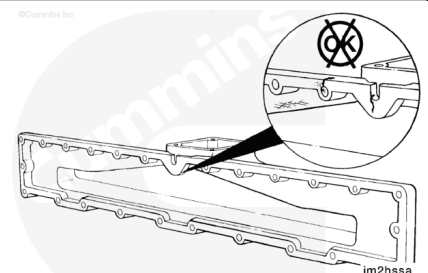
Clean the intake manifold gasket surface.



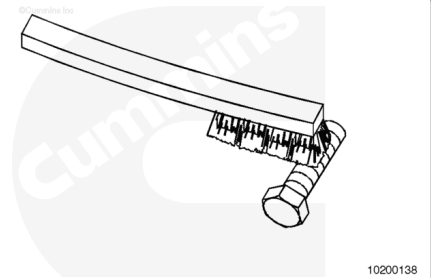
Inspect the air intake manifold for cracks or other damage.

Replace the air intake manifold if damage is found.

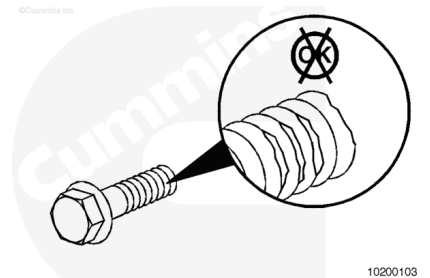
When inspecting the air intake manifold for oil or debris from an air system failure, also inspect the intake ports in the rocker lever housing and cylinder head for oil and debris.



Clean the intake manifold mounting capscrews.



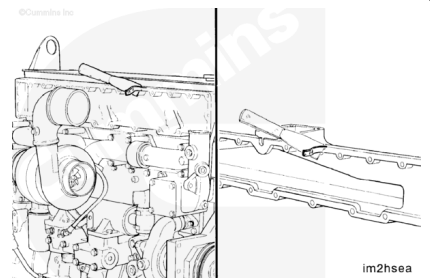
Inspect the air intake manifold mounting capscrews for stripped threads or other damage. Replace if damage is found.



Automotive with CM570

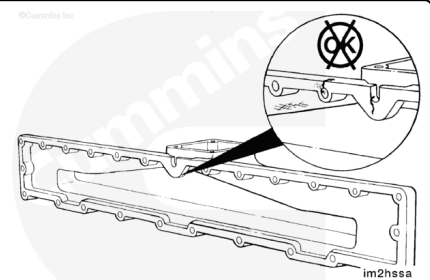
Clean the intake manifold to rocker housing surface.

Clean the intake manifold gasket surface.



Inspect the intake manifold for cracks or other damage.

Note : When inspecting the intake manifold for oil or debris from an air system failure, also inspect the intake ports in the rocker housing and cylinder head for oil debris.

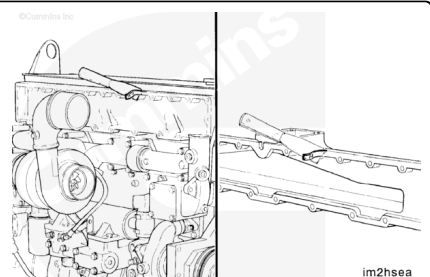


Automotive with CM876

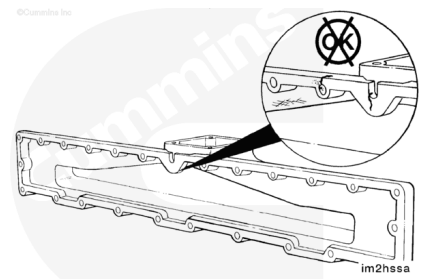
Clean the intake manifold gasket surface.

Inspect the air intake manifold and EGR valve mounting adapter for cracks or any other damage.

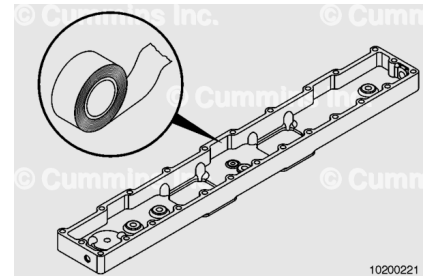
Replace the air intake manifold if damage is found.



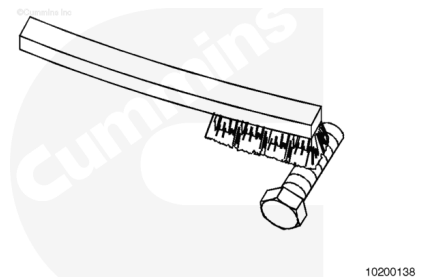
When inspecting the air intake manifold for oil or debris from the air system failure, also inspect the intake ports in the rocker lever housing and cylinder head for oil and debris.



After the component is clean and dry, cover open points with heavy tape.

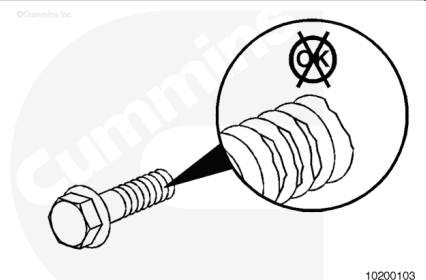


Clean the intake manifold mounting capscrews.



Inspect the air intake manifold mounting capscrews for stripped threads or any other damage

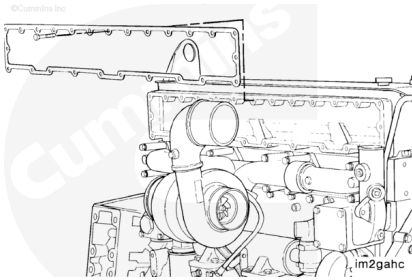
Replace the mounting capscrew if damage is found.



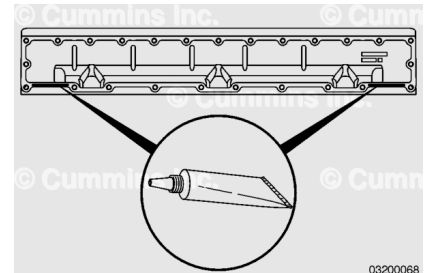
Install

Automotive with CM875, CM870 and CM570

Install two guide pins in the rocker lever housing before installing a new gasket.

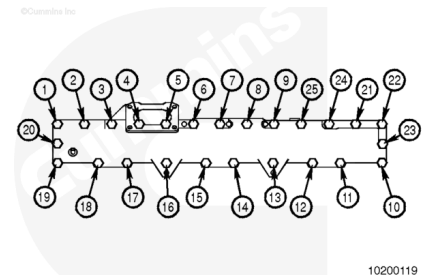


Apply a 2 mm [1/16 in] bead of sealant, Part Number 3823494 to rocker lever housing grooves if rocker lever housing is grooved.



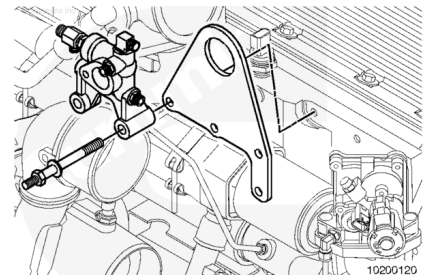
Install the air intake manifold over the guide pins. Install all of the capscrews except 21, 22, 23, 24, and 25. Finger tighten the capscrews.

Remove the guide pins.



Install the lifting bracket and the water manifold. Install capscrews 21, 22, 23, 24 and 25.

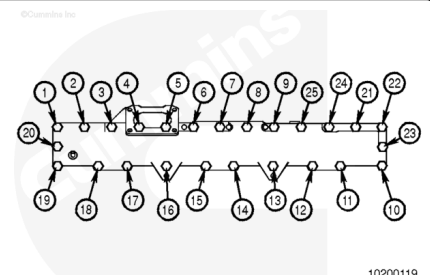
Finger tighten the capscrews.



Tighten all the capscrews in the sequence shown, except capscrews 13 and 16.

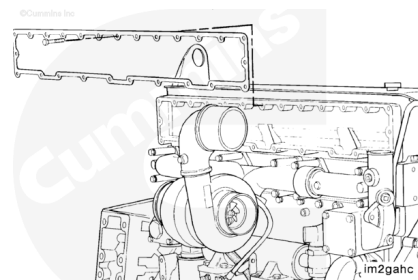
Tighten the two remaining capscrews.

Torque Value: 47 n•m [35 ft-lb]

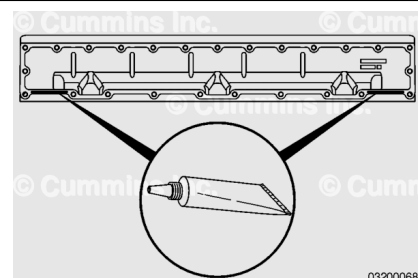


Automotive with CM570

Install two guide pins in the rocker lever housing before installing a new gasket.



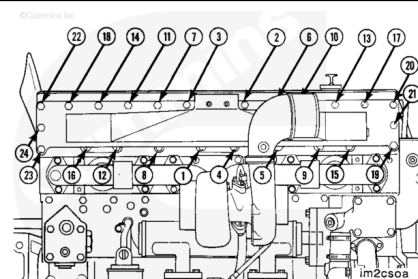
Apply a 2 mm [1/16 in] bead of sealant, Part Number 3823494, to rocker lever housing grooves if the rocker lever housing is grooved.



Install the intake manifold and capscrews.

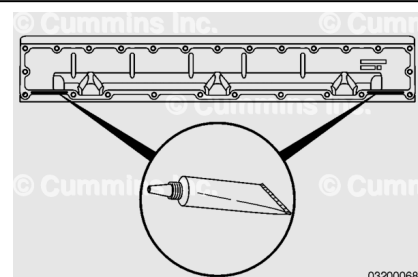
Tighten the capscrews in the sequence shown.

Torque Value: 47 n•m [35 ft-lb]



Automotive with CM876

Apply a 2 mm [1/16 in] bead of sealant, Part Number 3823494, to rocker lever housing grooves if the rocker lever housing is grooved.



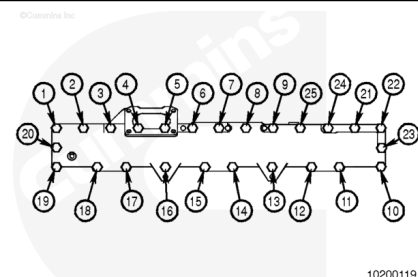
Install two guide pins in the rocker lever housing before installing a new gasket.

Install the air intake manifold over the guide pins.

Install all of the capscrews except 13, 16, 18, 19, 20, 21, 22, 23, 24, and 25.

Finger tighten the capscrews.

Remove the guide pins.



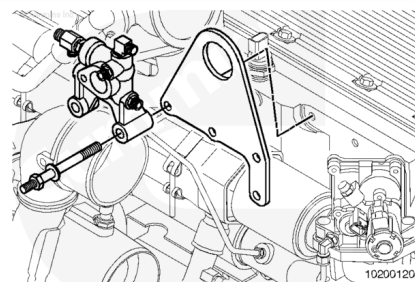
Install the lifting bracket and the water manifold.

Install capscrews 21, 22, 23, 24, and 25.

Finger tighten the capscrews.

Install the EGR valve and EGR valve bracket assembly and install capscrews 18, 19, and 20.

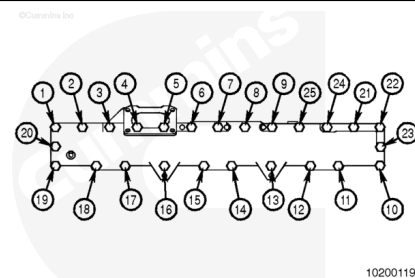
Finger tighten the capscrews. Refer to Procedure 011-022 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html>)



Tighten all the capscrews in the sequence shown, except capscrews 13 and 16.

Tighten the two remaining capscrews.

Torque Value: 47 n•m [35 ft•lb]



Finishing Steps

Automotive with CM875, CM870 and CM570

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Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the EGR cooler. Refer to Procedure 011-019 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-019-tr.html>)
- Install the water manifold return tube. Refer to Procedure 008-061 in Section 8.

(/qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html)

- Install the EGR valve. Refer to Procedure 011-022 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html)
- Install the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)
- Install the coolant vent line. Refer to Procedure 008-017 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-017-tr.html)
- Install the EGR valve coolant lines. Refer to Procedure 011-030 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-030-tr.html)
- Install the EGR differential pressure sensor adapter. Refer to Procedure 011-028 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-028-tr.html)
- Install the EGR differential pressure sensor. Refer to Procedure 019-370 in Section 19. (/qs3/pubsys2/xml/en/procedures/16/16-019-370.html)
- Install the exhaust gas pressure sensor. Refer to Procedure 019-376 in Section 19. (/qs3/pubsys2/xml/en/procedures/16/16-019-376.html)
- Install the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html)
- Install the intake manifold pressure sensor. Refer to Procedure 019-061 in Section 19. (/qs3/pubsys2/xml/en/procedures/82/82-019-061.html)
- Install the air intake connection to the air intake manifold and connect the intake manifold air temperature sensor connector to the engine harness. Refer to Procedure 010-080 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-080-tr.html)
- Install the EGR connection tube. Refer to Procedure 011-025 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html)
- Install the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11.

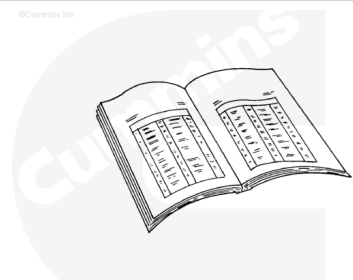
(/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html)

- Connect the EGR temperature sensor and intake manifold air temperature sensor connectors to the engine harness.
- Install the EGR mixer. Refer to Procedure 011-021 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-021-tr.html)
- Install the air compressor inlet tube to the EGR mixer. Refer to Procedure 012-109 in Section 12. (/qs3/pubsys2/xml/en/procedures/35/35-012-109-tr.html)
- Install the charge air cooler piping from the air intake adapter to the turbocharger compressor outlet. Refer to Procedure 010-027 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-027-tr.html)
- Install the air intake piping from the air cleaner to the turbocharger compressor inlet. Refer to the OEM service manual.
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Operate the engine and check for leaks.

Automotive with CM570

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



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Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing,

remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the inlet air connection and the sensors from the air manifold.
- Connect the air pipe from the charge air-cooler to the inlet air connection. Refer to the OEM service manual.
- Connect the battery. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Operate the engine and check for leaks.

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- Install the EGR cooler. Refer to Procedure 011-019 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-019-tr.html>)
- Install the aftertreatment fuel injector coolant return line. Refer to Procedure 011-051 in Section 11. (</qs3/pubsys2/xml/en/procedures/101/101-011-051-tr.html>)
- Install the water manifold and return tube. Refer to Procedure 008-061 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-061-tr.html>)
- Install the EGR cooler coolant lines. Refer to Procedure 011-031 in Section 11.



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(/qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html)

- Install the coolant vent line. Refer to Procedure 008-017 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-017-tr.html)
- Install the turbocharger and turbocharger actuator return lines. Refer to Procedure 010-041 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-041-tr.html)
- Install the EGR differential pressure sensor adapter. Refer to Procedure 011-028 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-028-tr.html)
- Install the EGR differential pressure sensor. Refer to Procedure 019-376 in Section 19. (/qs3/pubsys2/xml/en/procedures/16/16-019-376.html)
- Install the exhaust gas pressure sensor. Refer to Procedure 019-376 in Section 19. (/qs3/pubsys2/xml/en/procedures/16/16-019-376.html)
- Install the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html)
- Install the air intake connection. Refer to Procedure 010-080 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-080-tr.html)
- Install the EGR connection tube. Refer to Procedure 011-025 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html)
- Install the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html)
- Connect the EGR temperature sensor and intake manifold air pressure / temperature sensor connectors from the engine harness.
- Install the EGR mixer. Refer to Procedure 011-021 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-021-tr.html)
- Install the air compressor inlet tube from the EGR mixer. Refer to Procedure 012-109 in Section 12. (/qs3/pubsys2/xml/en/procedures/35/35-012-109-tr.html)

- Install the charge-air cooler piping from the air intake adapter to the turbocharger compressor outlet. Refer to Procedure 010-027 in Section 10.
(/qs3/pubsys2/xml/en/procedures/35/35-010-027-tr.html)
- Install the air intake piping from the air cleaner to the turbocharger compressor inlet. Refer to the OEM service manual.
- Connect the battery. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Fill the coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Operate the engine and check for leaks.